Experiment : 6

Date:

STATE CHART DIAGRAM

Aim

To design and understand the flowchart of a Library Book Issue System and to study the sequence of operations involved in issuing a book to a library member using logical decision making.

Objective

The objective of this experiment is to represent the workflow of a library book issuing process using a flowchart. This helps in understanding how the system verifies book availability, validates the member, checks borrowing limits, and finally issues the book while updating the system records.

Introduction

A library management system is used to manage the issuing and returning of books in a library. In such systems, it is important to maintain proper records of books and members to ensure smooth operation.

A flowchart is a graphical representation of a process that uses symbols to show the sequence of steps involved in solving a problem. Flowcharts help in understanding system logic easily and are widely used in system design and software engineering.

In this experiment, a flowchart is designed to represent the Library Book Issue System, which shows how a book request from a member is processed step by step.

System Description

The Library Book Issue System manages the process of issuing books to registered members. The system ensures that only valid members can borrow books and that the number of books issued does not exceed the allowed limit.

The system performs several checks such as book availability, member validation, and borrowing limits before issuing a book. These checks help maintain proper management of library resources.

Flowchart Explanation

1.Inquiry for Book

The process begins when a library member makes an inquiry about a particular book. The member requests the librarian to issue a book from the library.

2. Checking Book Availability

The librarian checks the availability of the requested book in the library database or catalog. This step determines whether the book is present in the library and ready to be issued.

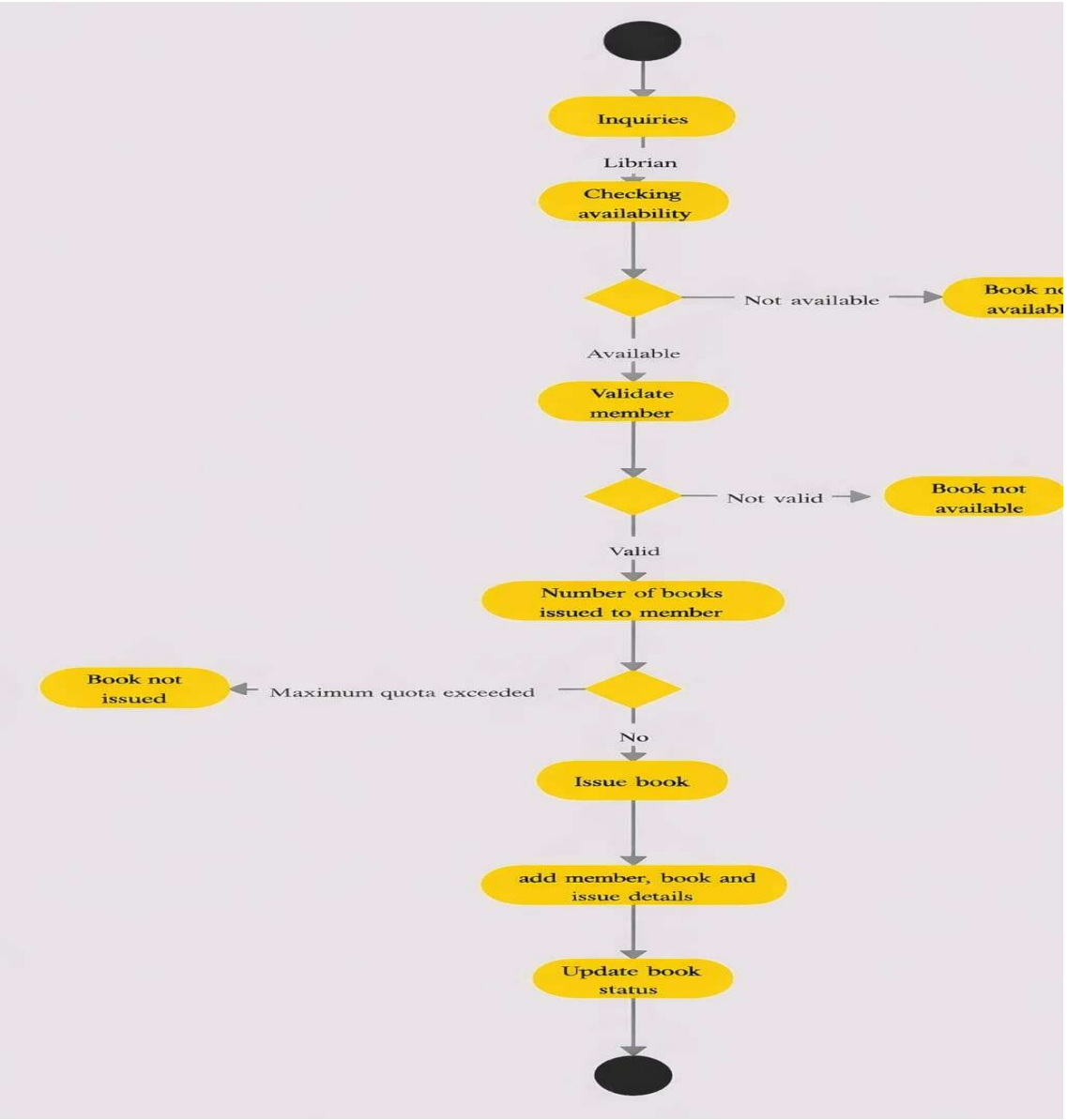
3. Book Availability Decision

If the book is not available, the system informs the member that the book is currently unavailable. The process stops at this point.

If the book is available, the system proceeds to the next step.

4. Member Validation

The system verifies the details of the member who requested the book. This step ensures that the member is properly registered in the library system



5. Valid Member Check

If the member is not valid or not registered, the system does not allow the book to be issued and the process stops.

If the member is valid, the system continues to the next step.

6. Checking Number of Books Issued

The system checks the number of books that are already issued to the member. Libraries usually have a maximum limit for the number of books a member can borrow at a time.

7. Maximum Quota Check

If the member has already reached the maximum borrowing limit, the system will not allow another book to be issued until a previously borrowed book is returned.

If the borrowing limit has not been exceeded, the process continues.

8. Issue Book

The librarian issues the requested book to the member.

9. Recording Issue Details

After issuing the book, the system records important details such as:

- Member ID
- Book ID
- Issue date
- Return date

These details are stored in the library database.

10. Updating Book Status

Finally, the status of the book is updated in the system to indicate that the book has been issued and is currently unavailable for other members.

Advantages of Using Flowchart

- Helps in understanding the process clearly.
- Makes system logic easy to follow.
- Identifies decision points in the process.
- Improves communication between developers and users.
- Helps in detecting logical errors during system design.

Applications

- Library Management Systems
- Student Management Systems
- Inventory Management Systems
- Banking and Transaction Systems
- Online Booking Systems

Result

Thus, the flowchart for the Library Book Issue System is designed successfully and the sequence of operations involved in issuing a book is clearly understood.